

8ai	Horizontal component of tension / spring force provides centripetal force Weight of sphere is (now) equal to the vertical component of tension / spring force OR horizontal and vertical components of tension / spring force combine to give a greater tension in spring Greater tension/spring force so <u>greater extension</u> / since extension is proportional to spring force	B1 M1 A1
8aii1	Radius, $r = 10.8 \sin 27^\circ = 4.903 \text{ cm}$ $\approx 4.9 \text{ cm}$	A1
8aii2	$F_{\text{spring}} \cos \theta = mg$ OR sum of vertical forces = 0 $F_{\text{spring}} = \frac{mg}{\cos \theta} = \frac{0.29 \times 9.81}{\cos 27^\circ} = 3.19 \text{ N}$ $F_{\text{spring}} \approx 3.2 \text{ N}$ (shown)	B1 A1 A0
8aii3	$k = \frac{\Delta F_{\text{spring}}}{\Delta x} = \frac{3.2 - (0.29 \times 9.81)}{(10.8 - 8.5)}$ $k = 0.15 \text{ N cm}^{-1}$	C1 A1
8aiii 1	$a_c = \frac{F_{\text{spring}} \sin \theta}{m} = \frac{3.2 \times \sin 27^\circ}{0.29}$ $a_c = 5.0 \text{ m s}^{-2}$	C1 A1
8aiii 2	$a_c = r \omega^2 = r \left(\frac{2\pi}{T} \right)^2$ $T = 2\pi \times \sqrt{(0.049 / 5.0)} = 0.62 \text{ s}$	C1 A1

8bi	From $g = -\frac{d\phi}{dr}$, when the gravitational field strength is zero, the potential gradient at that point would be zero. (Thus the point will be a turning point, which is a maximum point.) Thus, x is 0.52×10^{12} m Accepted range of x : 0.50×10^{12} m – 0.54×10^{12} m	M1 A1
8bii1	$\Delta U = m\Delta\phi = 180 \times [-14 - (-10)] \times 10^8$ $= -7.2 \times 10^{10}$ J Change = 7.2×10^{10} J The change in kinetic energy is an <u>increase</u>.	C1 A1 A1
8bii2	energy required (to reach maximum point) = $180 \times (10 - 4.4) \times 10^8$ or energy per unit mass (to reach maximum point) = $(10 - 4.4) \times 10^8$ $\frac{1}{2} \times 180 \times v^2 = 180 \times (10 - 4.4) \times 10^8$ or $\frac{1}{2} \times v^2 = (10 - 4.4) \times 10^8$ $v = 3.3 \times 10^4$ m s⁻¹	C1 C1 A1